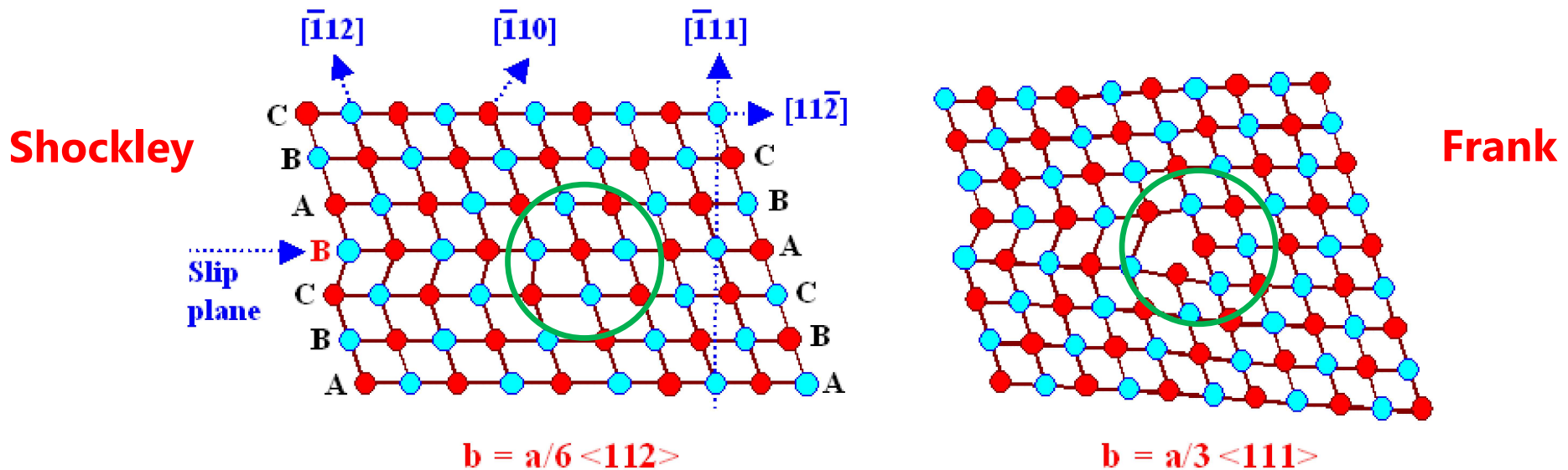


2.5.3 Partial dislocations and stacking faults

- The **displacement vector** is $\mathbf{b} = \mathbf{a}/6\langle 112 \rangle$
- Dislocations with this type of Burgers vector are called **Shockley partial dislocations**
- In the $\langle 110 \rangle$ projection, Shockley and Frank partial dislocations appear as shown below



These dislocations are more or less the only type of dislocations that really exist in fcc crystals (because perfect dislocations of type $\mathbf{a}/2\langle 110 \rangle$ split into partial dislocations $\mathbf{a}/6\langle 112 \rangle + \mathbf{a}/3\langle 11\bar{1} \rangle$)